

Project Design Phase

Problem - Solution Fit

Date	October 03,2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL: Restrict Record Access Based On Field Value
Maximum Marks	4 Marks

Solution Architecture Definition

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its core functional goals within this architecture are to:

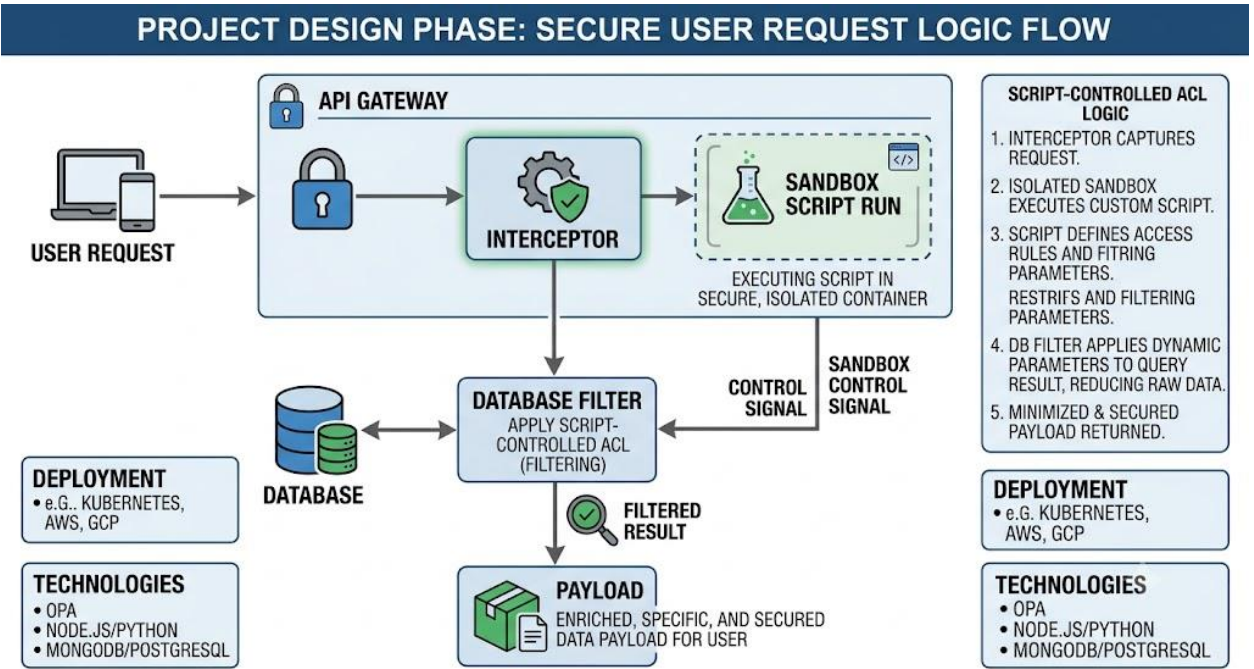
- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define architectural blocks, execution phases, and complex solution requirements.
- Provide technical specifications according to which the solution is defined, managed, and delivered accurately.

Proposed Architectural Framework Diagram Description

To solve the issue of broad exposures or static role limits, our Script-Controlled Access Control List (ACL) framework operates across a decoupled multi-tiered pipeline. This structure ensures granular isolation by inspecting row criteria in real-time.

Layer Component	Technical Workflow / Responsibilities	Technology Block
1. Presentation / Administration UI	Provides security admin console screens where operators create rules, track violations, and write customized validation conditional scripts.	ReactJS / Tailwind CSS UI
2. API Gateway Interceptor Middleware	Intercepts inbound data transactions. Fetches current session identifiers and contextual field values from target rows prior to rendering.	Node.js Express / Python FastAPI
3. Sandboxed Logic Scripting Engine	Isolates execution scripts from core process loops. Safely processes custom constraints against field values (e.g., Status=='Confidential') to output true/false choices.	Isolated-VM / PySandbox Hub
4. Database Storage Core	Holds structural transaction tables alongside independent ACL validation logic scripts matrices and secure access tracks.	PostgreSQL / IBM DB2 Layer

5. Cryptographic Logging Hub	Asynchronously records validation audit records, tracking blocked attempts or dynamic policy evaluations for compliance verification.	Redis Cache / Winston Log Core
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Data Interception and Evaluation Characteristics

1. Query Extraction: When an end-user triggers a read request, the API interceptor captures the contextual identifiers.
2. Sandboxed Verification: The validation controller loads script rules matched against specific record states.
3. Secure Payload Formatting: Records failing validation criteria checks are excluded from datasets instantly before payload response dispatching.